Label-Efficient Hyperspectral Image Change Detection via Multiscale Self-Distillation

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***Abstract*—Methods based on deep learning have achieved remarkable success in hyperspectral image change detection (HSI-CD). However, the high dependence on massive labeled data and the computational complexity of state-of-the-art models pose significant challenges, particularly in scenarios with extreme label scarcity. To address these issues, a multiscale self-distillation change detection (MSDistillCD) framework is proposed. In the proposed framework, two core modules are designed to extract robust features and enforce consistency under limited supervision. First, a multiscale feature extraction module (MSFEM) constructs subpatches aligned at the center from a fixed maximum patch and generates inputs with explicit difference information. A shared convolutional neural network (CNN) backbone with dual global aggregation is then employed to capture embeddings that remain robust across scales. Subsequently, a self-distillation classification module (SDCM) is utilized to facilitate knowledge transfer within the network. Unlike conventional distillation settings, the smallest-scale branch is treated as the primary branch and guides the larger-scale auxiliary branches through temperature-scaled self-distillation, providing additional regularization during training while allowing efficient inference with a single branch. Experimental results on benchmark HSI datasets demonstrate that MSDistillCD achieves the best overall accuracy (OA), $\kappa \times 100$, and $F1$ across all three datasets, while maintaining robust performance even with extremely limited training samples (e.g., 0.1%). The source code of the proposed MSDistillCD will be released at https://github.com/zhangyuan1698-sketch/MSDistillCD**

***Index Terms*—Dual global aggregation, hyperspectral image change detection (HSI-CD), multiscale feature extraction, self-distillation.**

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I. Introduction

HYPERSPECTRAL image change detection (HSI-CD) is an important task in Earth observation and is widely used in precision agriculture, urban planning, environmental monitoring, and postdisaster assessment [1], [2]. The core objective of HSI-CD is to detect land-cover or surface changes by comparing bitemporal hyperspectral observations collected from the same scene at different acquisition dates. Unlike traditional optical imagery, hyperspectral sensors capture hundreds of continuous spectral bands, offering rich diagnostic information that is often difficult to capture with multispectral data [3], [4]. Consequently, using this rich spectral–spatial information in HSIs for accurate change interpretation has become a primary focus in the remote sensing field. More broadly, hyperspectral learning has also been actively studied in RGB-to-HSI reconstruction, object tracking, and marine oil spill detection [5], [6], [7].

In the early stages of development, HSI-CD relied heavily on algebra-based and statistical methods. Classical approaches such as change vector analysis (CVA) [8], multivariate alteration detection (MAD) [9], and principal component analysis with K-means clustering (PCAKmeans) [10] served as foundational baselines [11], [12]. These methods typically generate difference images followed by thresholding techniques or clustering algorithms to obtain binary change maps [13]. While computationally efficient, these traditional methods suffer from significant limitations. They rely on manually designed features, are highly sensitive to radiometric noise, and often struggle to model the complex, nonlinear spectral relationships inherent in hyperspectral data with many dimensions, leading to suboptimal performance in complex scenarios.

To overcome the limitations of shallow models, convolutional neural networks (CNNs) have become a dominant paradigm for HSI-CD because they can jointly learn spatial–spectral representations and offer powerful feature extraction capabilities. Representative CNN-based studies have explored two-stream encoder–decoder architectures with contextual fusion, multiscale pyramid representations, and lightweight spatial–spectral modeling [3], [14], [15]. Siamese and dual-branch CNN variants have also been widely investigated, among which GTMSiam [16] is a representative example for learning change-aware bitemporal features. Related hyperspectral analysis studies have also explored efficient CNNs and graph-enhanced CNN models [17], [18].

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However, for HSI-CD, the prediction target is the center pixel, whereas the evidence for that decision may reside at multiple surrounding spatial extents. Existing CNN designs do not explicitly resolve this mismatch between supervision focused on the center and context-dependent representation learning.

Transformer-based and attention-based methods have attracted increasing interest in recent years for HSI-CD, largely due to their strong capability of modeling long-range dependencies and global contextual relationships in spectral–spatial and temporal domains. Representative architectures include spectral–spatial–temporal transformers [1], [19], [20] and more efficiency-oriented attention designs such as GLAFormer [21]. Related hyperspectral analysis works have also investigated hybrid CNN-Transformer and FourierKAN-Transformer designs [22], [23]. While these models improve global context modeling, they typically remain parameter-heavy and computationally intensive, often requiring substantial labeled data to converge without overfitting.

To alleviate the issue caused by limited samples in HSI-CD, existing studies have explored a variety of label-efficient strategies, including transfer learning and pretraining paradigms, data augmentation, pseudo-labeling, and more complex semi-supervised and self-supervised learning frameworks. Related studies have also explored few-shot, small-sample, domain-aware augmentation, and progressive graph learning under limited supervision [24], [25], [26], [27]. In semi-supervised learning, Luo et al. [28] proposed DCENet to jointly leverage labeled and unlabeled samples via a diff-feature contrast enhancement scheme with multiscale divergence regularization and probabilistic contrastive objectives, while Qu et al. [29] developed MSCSCNet by unfolding multiscale convolutional sparse coding into a semi-supervised teacher–student framework, improving interpretability and utilizing limited supervision. Beyond schemes built on consistency regularization, pseudo-label-driven augmentation is also widely used. For example, Jiang et al. [30] proposed ACFHAN, which generates supervisory signals with high confidence through multiscale augmentation based on superpixels to expand training data under weak labeling. Meanwhile, transfer learning and self-supervised pretraining have become popular alternatives to reduce annotation demand, and generative pretraining has also been explored, e.g., GPSST [31] uses masked autoencoding on large HSI patch collections before semi-supervised CD. Despite their progress, these approaches often introduce new limitations. First, pretraining and hybrid semi-supervised pipelines can be computationally heavy and cross-domain sensitive [31]. Second, pseudo-label or augmentation based on superpixels may accumulate noise and rely on handcrafted procedures [30]. Besides, many semi-supervised and self-supervised methods are validated under conventional low-label ratios (e.g., 1%–5%) rather than extreme budgets such as 0.1% per scene [28], [29]. Consequently, there remains a gap between the goal of truly lightweight, robust, and effective under extreme label scarcity in HSI-CD and the practical complexity or supervision assumptions of existing solutions.

To address the above challenges, we propose multiscale self-distillation change detection (MSDistillCD), a label-efficient and computationally efficient framework for bitemporal HSI-CD. The framework consists of two components, a multiscale feature extraction module (MSFEM) and a self-distillation classification module (SDCM). MSFEM is designed to build multiscale views aligned at the center from a fixed patch and enhances them with explicit cues about change, enabling the network to capture both local and nearby contextual evidence under limited supervision. Based on these multiscale representations, SDCM adopts a lightweight self-distillation strategy in which the smallest-scale branch serves as the primary branch and regularizes the larger-scale auxiliary branches during training, while only the primary branch is retained at inference time. In this way, MSDistillCD makes use of multiscale information during training while preserving the efficiency of a lightweight CNN pipeline. The core contributions of this article can be summarized as follows.

1) Aiming at the issues of poor feature representation capability and performance degradation under extreme label scarcity in existing HSI-CD methods, we propose an end-to-end MSDistillCD framework. It integrates explicit difference-aware modeling and multiscale self-distillation learning, effectively improving detection accuracy and generalization ability in scarce annotation scenarios.
2) To alleviate the limitation that conventional patch-based methods cannot capture stable scale-robust features and lack effective change priors, we design the MSFEM module. It constructs center-aligned multiscale views from a fixed maximum patch and employs a shared CNN combined with dual pooling aggregation, yielding discriminative representations that maintain stable characterization across different scales.
3) To tackle the problems of heavy inference overhead and insufficient multiscale feature consistency learning in existing semi-supervised schemes, we develop a lightweight internal self-distillation module, SDCM. It realizes intra-network knowledge transfer to enhance multiscale feature consistency under extremely limited labels, and maintains high inference efficiency by deploying only the primary branch in the testing phase.

The remainder of this article is organized as follows. Section II details the proposed MSDistillCD. Section III presents experimental results and ablation analyses. Section IV concludes this article and discusses future research directions.

II. Methodology

A. HSI-CD Formulation and Framework Overview

Let $\mathbf{I}^{(1)}, \mathbf{I}^{(2)} \in \mathbb{R}^{C\times H\times W}$ denote two co-registered hyperspectral images acquired over the same scene at two different times, where H and W are the spatial dimensions and C is the number of spectral bands. The goal of HSI-CD is to determine whether each spatial location has changed between the two acquisition times. Following common learning settings based on patches, for each labeled center pixel, we extract a pair of patches from the two acquisition times:

$$\mathbf{X}_1, \mathbf{X}_2 \in \mathbb{R}^{C\times P\times P} \tag{1}$$

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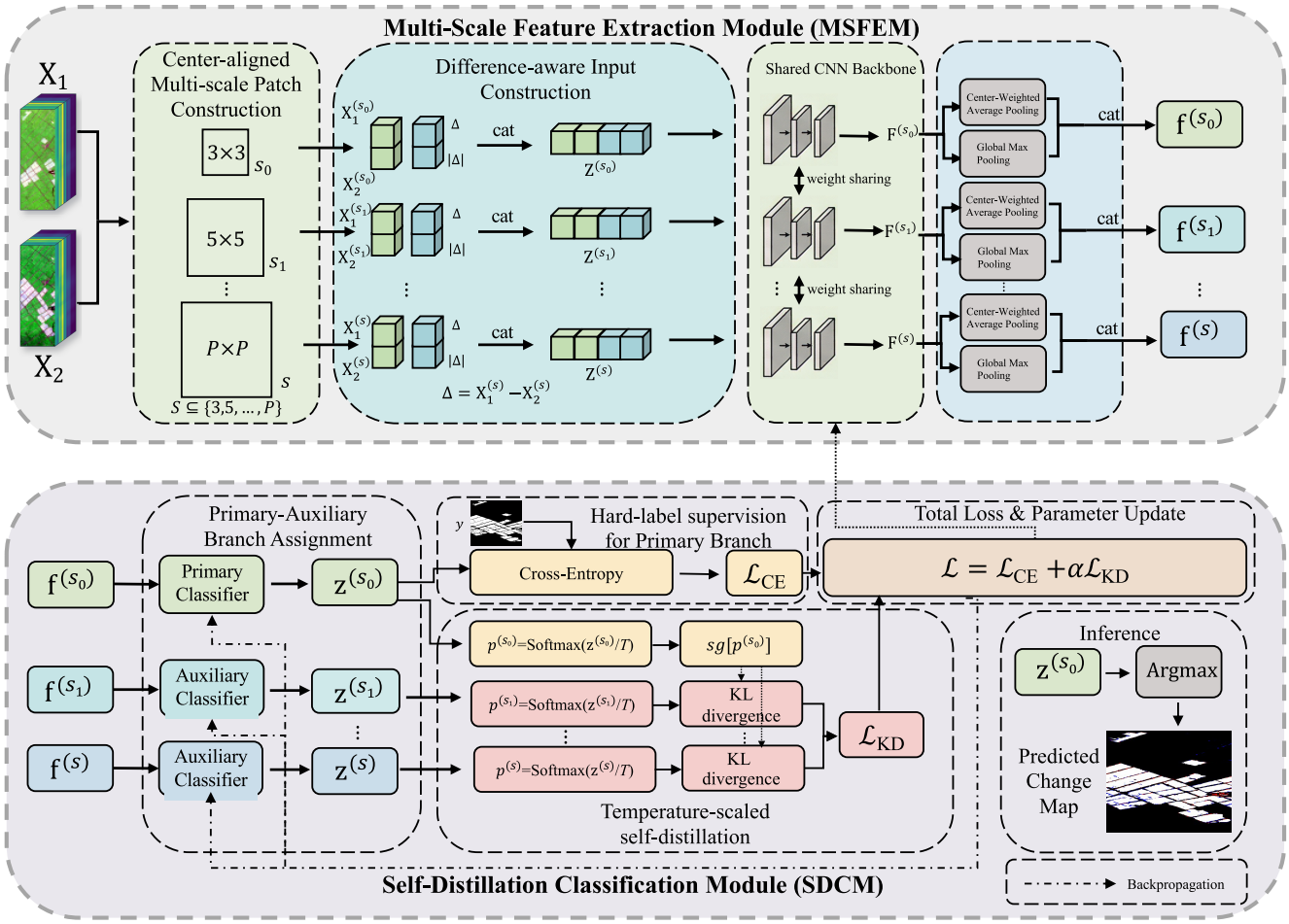


Fig. 1. Overall architecture of MSDistillCD. MSFEM constructs multiscale inputs aligned at the center and enriched with explicit difference information, and extracts discriminative embeddings by using a shared lightweight CNN backbone with dual global aggregation. Based on these embeddings, SDCM performs self-distillation from the primary branch to the auxiliary branches during training and retains only the primary branch for efficient inference.

where P is the maximum patch size, and the label is denoted by $y \in \{0, 1\}$, with 0 and 1 representing unchanged and changed classes, respectively.

The proposed MSDistillCD framework consists of two components: MSFEM and SDCM, as illustrated in Fig. 1. Given a pair of patches from the two acquisition times $\{\mathbf{X}_1, \mathbf{X}_2\}$, MSFEM first constructs multiple subpatches aligned at the center at different spatial scales and transforms each scale into a representation with explicit difference information. These multiscale inputs are then fed into a shared lightweight CNN backbone to extract discriminative embeddings. Subsequently, SDCM treats the branch at the smallest scale as the primary branch and the remaining branches at larger scales as auxiliary branches. During training, the primary branch is directly supervised by true labels, while the auxiliary branches are optimized to match the softened output distribution of the primary branch through temperature-scaled distillation. During inference, only the primary branch is retained, yielding an efficient prediction pipeline with the runtime of a model with a single branch.

To simplify notation, all quantities associated with different scales are indexed by the spatial scale itself. Specifically, let $\mathcal{S}$ denote the set of selected spatial scales, let $s \in \mathcal{S}$ denote an arbitrary scale, and let $s_0 = \min\{\mathcal{S}\}$ denote the smallest scale, which is used as the primary branch.

B. Multiscale Feature Extraction Module

1) Patch Construction Across Multiple Scales: Given an odd maximum patch size $P \times P$, we define a set of valid spatial scales

$$\mathcal{S} \subseteq \{3, 5, \ldots, P\} \tag{2}$$

where each element in $\mathcal{S}$ is an odd integer. The smallest scale $s_0 = \min\{\mathcal{S}\}$ is used as the primary branch.

For each scale $s \in \mathcal{S}$, a center crop operator $\mathcal{C}_s(\cdot)$ is applied to obtain the corresponding subpatches

$$\mathbf{X}_1^{(s)} = \mathcal{C}_s(\mathbf{X}_1), \qquad \mathbf{X}_2^{(s)} = \mathcal{C}_s(\mathbf{X}_2) \tag{3}$$

where

$$\mathbf{X}_1^{(s)}, \mathbf{X}_2^{(s)} \in \mathbb{R}^{C \times s \times s}. \tag{4}$$

This design enables the network to observe the same center pixel under different receptive field sizes, thereby incorporating both local and nearby contextual evidence without introducing multiple independent backbones.

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2) Difference-Aware Input Construction: To explicitly encode temporal discrepancy information, for each scale $s \in \mathcal{S}$, the raw patches from the two acquisition times and their difference maps are jointly used. Specifically, the signed difference and absolute difference are computed as

$$\mathbf{D}^{(s)} = \mathbf{X}_1^{(s)} - \mathbf{X}_2^{(s)} \tag{5}$$

$$|\mathbf{D}^{(s)}| = |\mathbf{X}_1^{(s)} - \mathbf{X}_2^{(s)}|. \tag{6}$$

The final input tensor for scale s is formed by concatenation along the spectral dimension, where Cat[·] denotes this operation

$$\mathbf{Z}^{(s)} = \mathrm{Cat}\left[\mathbf{X}_1^{(s)}, \mathbf{X}_2^{(s)}, \mathbf{D}^{(s)}, |\mathbf{D}^{(s)}|\right] \in \mathbb{R}^{4C\times s\times s}. \tag{7}$$

Compared with using only the raw pair from the two acquisition times, this explicit representation of differences provides the network with both original spectral–spatial observations and direct change cues, which is particularly beneficial in scenarios with limited samples where learning implicit differencing from limited supervision is difficult.

3) Shared CNN Backbone: All inputs from different scales share the same convolutional feature extractor. Let $\Phi(\cdot;\theta)$ denote the shared backbone parameterized by θ. The backbone is composed of a stack of lightweight convolutional blocks, each consisting of a 3×3 convolution, batch normalization, and a ReLU activation.

Given an input tensor $\mathbf{Z}^{(s)}$, the backbone first maps it to a hidden representation with d bands through a stem convolutional block, followed by several additional convolutional blocks with the same spectral dimension. Since all convolutions use a stride of 1 and a padding of 1, the spatial resolution at each scale is preserved throughout feature extraction. The resulting feature map is denoted by

$$\mathbf{F}^{(s)} = \Phi\left(\mathbf{Z}^{(s)};\theta\right) \in \mathbb{R}^{d\times s\times s}. \tag{8}$$

The shared backbone parameters θ are randomly initialized and optimized end-to-end during training. The use of a shared backbone has two advantages. First, it keeps the total number of learnable parameters small, which is important under extreme label scarcity. Second, it forces different scales to be embedded in a common feature space, facilitating effective knowledge transfer in the subsequent self-distillation stage.

4) Dual Global Aggregation: After convolutional feature extraction, each feature map from a given scale is aggregated into a compact feature vector through two complementary global operations: average pooling with greater emphasis on the center and global max pooling.

To better align learning on patches with supervision of the center pixel, we introduce a pooling strategy that places greater emphasis on the center. For a feature map of spatial size $s \times s$ and $i, j = 1, \ldots, s$, the normalized coordinates are defined on a symmetric grid over $[-1, 1]$ as

$$u_i = \frac{2(i-1)}{s-1} - 1, \qquad v_j = \frac{2(j-1)}{s-1} - 1. \tag{9}$$

The Gaussian weight at location (i, j) is given by

$$\omega_{ij}^{(s)} = \frac{\exp\left(-\frac{u_i^2+v_j^2}{2\sigma^2}\right)}{\sum_{p=1}^{s}\sum_{q=1}^{s}\exp\left(-\frac{u_p^2+v_q^2}{2\sigma^2}\right)} \tag{10}$$

where $u_i, v_j \in [-1, 1]$ are the normalized spatial coordinates and σ controls the concentration around the patch center. In our implementation, σ is set to 0.5. The weighted average feature with greater emphasis on the center is then computed as

$$\mathbf{f}_{\mathrm{avg}}^{(s)}(c) = \sum_{i=1}^{s}\sum_{j=1}^{s}\omega_{ij}^{(s)}\mathbf{F}^{(s)}(c,i,j), \qquad c = 1, \ldots, d. \tag{11}$$

In parallel, a global max pooling operation is applied to capture the most salient activation response

$$\mathbf{f}_{\mathrm{max}}^{(s)}(c) = \max_{1\le i\le s,\ 1\le j\le s}\mathbf{F}^{(s)}(c,i,j). \tag{12}$$

The final representation in the center pixel of a patch for scale s is obtained by concatenating the two pooled vectors

$$\mathbf{f}^{(s)} = \mathrm{Cat}\left[\mathbf{f}_{\mathrm{avg}}^{(s)}, \mathbf{f}_{\mathrm{max}}^{(s)}\right] \in \mathbb{R}^{2d}. \tag{13}$$

This dual aggregation mechanism provides a balance between contextual summarization that emphasizes the center and strong response preservation, yielding robust and discriminative embeddings for subsequent classification.

C. Self-Distillation Classification Module

1) Motivation for Self-Distillation: Although MSFEM provides representations from multiple spatial scales, directly supervising all scale branches only with scarce hard labels can lead to unstable and inconsistent predictions across scales. This issue is particularly important in HSI-CD under extremely limited annotation, where larger patches may introduce useful contextual information but may also contain irrelevant background pixels or mixed boundary regions. Since the class label is assigned to the center pixel, the branch with the smallest scale is more tightly aligned with the prediction target and provides the most direct local decision reference. Therefore, self-distillation is introduced so that the primary branch at the smallest scale can regularize the auxiliary branches at larger scales. In this way, the model can use additional spatial context during training while keeping the decision behavior consistent with the supervision on the center pixel. Moreover, the distillation is performed within the same network and requires no external model or additional branch during inference, which is essential for maintaining a lightweight inference pipeline.

2) Primary and Auxiliary Branches: After feature extraction, each embedding from a given scale $\mathbf{f}^{(s)}$ is fed into a linear classifier for that scale. The smallest-scale branch is used as the primary branch

$$\mathbf{z}^{(s_0)} = \mathbf{W}_{s_0}\mathbf{f}^{(s_0)} + \mathbf{b}_{s_0} \tag{14}$$

where $\mathbf{z}^{(s_0)} \in \mathbb{R}^M$ denotes the logits of the primary branch, $\mathbf{W}_{s_0} \in \mathbb{R}^{M\times 2d}$ and $\mathbf{b}_{s_0} \in \mathbb{R}^M$ are the weight matrix and bias vector of the primary classifier, and M is the number of classes. In this work, $M = 2$ for binary classification.

For each remaining scale $s \in \mathcal{S} \setminus \{s_0\}$, an independent auxiliary classifier is used

$$\mathbf{z}^{(s)} = \mathbf{W}_s\mathbf{f}^{(s)} + \mathbf{b}_s \tag{15}$$

where $\mathbf{W}_s \in \mathbb{R}^{M\times 2d}$ and $\mathbf{b}_s \in \mathbb{R}^M$ denote the weight matrix and bias vector of the classifier at scale s. Thus, the smallest

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receptive field acts as the primary branch, while the remaining larger receptive fields act as auxiliary branches.

This design differs from conventional distillation strategies, where the source of soft targets is often a deeper or larger model. In HSI-CD, however, the supervision target corresponds to the center pixel, and the smallest patch is more closely tied to the local evidence around that pixel. Here, primary refers to the branch used for final prediction rather than to model size. Therefore, using the smallest-scale branch as the primary branch provides a suitable reference for regularizing the larger-scale auxiliary branches under extremely limited labels.

3) Hard-Label Supervision for the Primary Branch: The primary branch is directly optimized with the standard cross-entropy loss

$$\mathcal{L}_{\mathrm{CE}} = \mathrm{CE}\left(\mathbf{z}^{(s_0)}, y\right) \tag{16}$$

where $\mathrm{CE}(\cdot,\cdot)$ denotes the cross-entropy loss and y is the true label. This term ensures that the smallest-scale branch serves as the primary discriminative predictor.

4) Temperature-Scaled Self-Distillation: To transfer knowledge from the primary branch to the auxiliary branches, the predictive distribution of the primary branch is softened with a temperature parameter $T > 0$

$$\mathbf{p}^{(s_0)} = \mathrm{Softmax}\left(\frac{\mathbf{z}^{(s_0)}}{T}\right). \tag{17}$$

Similarly, for each auxiliary scale $s \in \mathcal{S} \setminus \{s_0\}$, the softened distribution is

$$\mathbf{p}^{(s)} = \mathrm{Softmax}\left(\frac{\mathbf{z}^{(s)}}{T}\right). \tag{18}$$

A larger temperature T produces a softer class distribution, exposing information about similarities between classes that facilitates knowledge transfer from the primary branch to the auxiliary branches [32]. The knowledge distillation (KD) loss between the primary branch and the auxiliary branch at scale s is defined as the Kullback–Leibler (KL) divergence [33]

$$\mathcal{L}_{\mathrm{KD}}^{(s)} = T^2\, D_{\mathrm{KL}}\left(\mathrm{sg}\left[\mathbf{p}^{(s_0)}\right] \,\|\, \mathbf{p}^{(s)}\right) \tag{19}$$

where $D_{\mathrm{KL}}(P\|Q) = \sum_i P(i)\log P(i)/Q(i)$, and $\mathrm{sg}[\cdot]$ denotes the stop-gradient operator. In the distillation branch, the primary distribution is detached; therefore, the primary branch is not updated by distillation gradients and is learned only from hard-label supervision.

The overall distillation loss is computed by averaging across all auxiliary branches

$$\mathcal{L}_{\mathrm{KD}} = \frac{1}{|\mathcal{S}| - 1} \sum_{s \in \mathcal{S}\setminus\{s_0\}} \mathcal{L}_{\mathrm{KD}}^{(s)}. \tag{20}$$

5) Overall Objective Function: The final training objective is the weighted sum of the primary cross-entropy loss and the self-distillation loss

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \alpha \mathcal{L}_{\mathrm{KD}} \tag{21}$$

where $\alpha \in [0, 1]$ is a hyperparameter that balances hard-label supervision and distillation across scales. The temperature T controls the smoothness of the primary and auxiliary probability distributions, while α determines the strength of the distillation regularization. In this article, α and T are treated as fixed hyperparameters in the main comparison experiments, and their effects are analyzed separately in the parameter sensitivity study.

Although the primary distribution is detached in the distillation branch, the backbone parameters are shared across all scales. Therefore, the shared feature extractor is still jointly optimized by the primary cross-entropy loss and the auxiliary distillation losses through backpropagation. During training, θ is updated by minimizing the total loss

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L} \tag{22}$$

where η denotes the learning rate. As a result, the shared feature extractor is jointly shaped by direct supervision from the primary branch and consistency constraints from the auxiliary branches, which improves representation robustness without increasing inference complexity.

D. Training and Inference Strategy

During training, the network processes all scales in parallel. For each scale, MSFEM generates a difference-aware input, the shared backbone extracts a compact representation, and SDCM produces one primary prediction and multiple auxiliary predictions. The optimization objective enforces both label fidelity and multiscale prediction consistency. This training strategy enables the model to exploit multiscale context even when only a very limited number of labeled samples are available.

During inference, only the primary branch associated with the smallest scale is used

$$\hat{y} = \arg\max\left(\mathbf{z}^{(s_0)}\right) \tag{23}$$

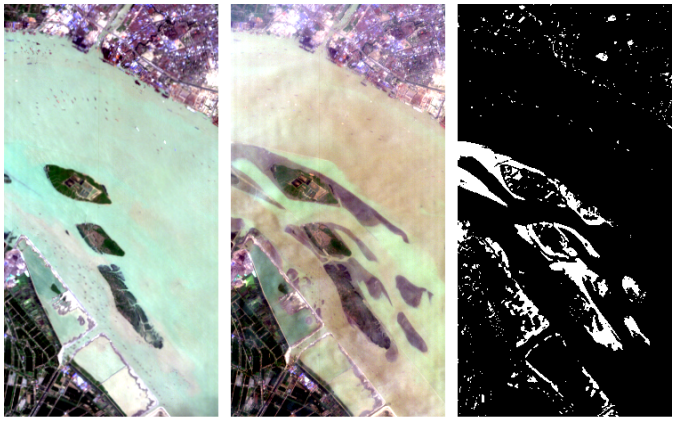
where $\hat{y}$ denotes the predicted class label. Consequently, MSDistillCD preserves the efficiency of a lightweight CNN with a single branch at test time, while benefiting from multiscale feature learning and self-distillation during training.

E. Discussion of Framework Design

The proposed design is motivated by the practical challenges of HSI-CD under extreme label scarcity. First, multiscale cropping aligned at the center allows the model to observe each sample under different spatial contexts without introducing multiple independent encoders. Second, explicit concatenation of raw patches from the two acquisition times, signed differences, and absolute differences provides strong priors about change that reduce the burden of representation learning from scarce labels. Third, the shared CNN backbone with dual global aggregation produces compact yet expressive embeddings by combining contextual summarization that emphasizes the center and salient activation preservation. Finally, the lightweight self-distillation mechanism regularizes branches with additional context to remain consistent with the primary branch, which is more closely tied to the center label. This helps stabilize learning while maintaining efficient inference.

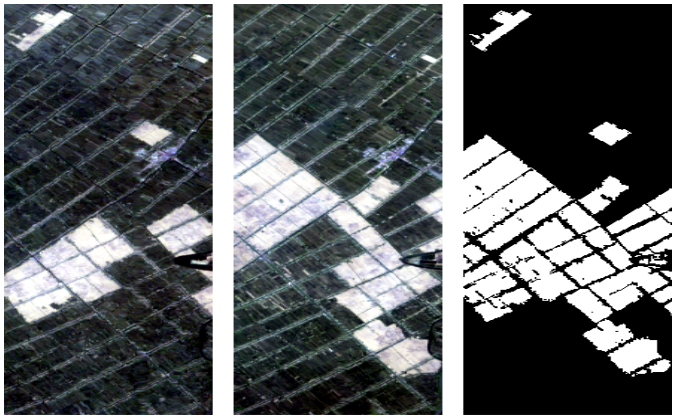
Overall, MSDistillCD offers a simple, end-to-end, and computationally efficient solution for HSI-CD, making it particularly suitable for practical remote sensing scenarios where annotation resources are severely limited.

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(a) (b) (c)

Fig. 2. River dataset. (a) Imagery acquired on May 3, 2013. (b) Imagery acquired on December 31, 2013. (c) Ground truth.



(a) (b) (c)

Fig. 3. Farmland dataset. (a) Imagery acquired on May 3, 2006. (b) Imagery acquired on April 23, 2007. (c) Ground truth.

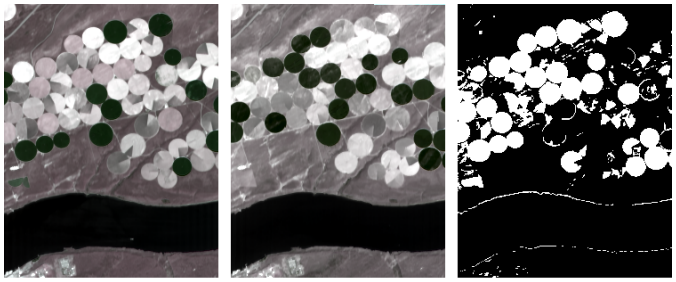
III. Experimental Results and Analysis

This section presents the experimental protocol, comparison results, and ablation studies of MSDistillCD on benchmark HSI-CD datasets.

A. Datasets and Evaluation Metrics

We conduct experiments on three benchmark bitemporal HSI-CD datasets, namely River, Farmland, and USA. The acquisition dates, geographical locations, sensor, spatial and spectral resolutions, preprocessing, image sizes, retained bands, and class distributions are summarized in Table I. All three datasets were acquired by the EO-1 Hyperion sensor and preprocessed by removing noisy and water absorption bands. Their RGB composites and ground truth maps are shown in Figs. 2–4.

These datasets exhibit different change patterns and spatial structures. River contains fragmented changes along the river channel, Farmland mainly includes relatively regular changes in cultivated areas and crop cover, and USA involves agricultural and riverbank variations. Therefore, they provide a representative testbed for evaluating HSI-CD methods under limited labeled data.



(a) (b) (c)

Fig. 4. USA dataset. (a) Imagery acquired on May 1, 2004. (b) Imagery acquired on May 8, 2007. (c) Ground truth.

To quantitatively evaluate detection performance, we adopt five widely used metrics, namely overall accuracy (OA), Kappa coefficient ($\kappa \times 100$), $F1$-score ($F1$), precision (P), and recall (R). Let TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively, and let $N = \mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}$. Then

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{N}, \quad P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$
$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \quad \mathrm{F1} = \frac{2P \times R}{P + R}. \tag{24}$$

The Kappa coefficient is computed as

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{25}$$

where $p_o = \mathrm{OA}$ and p_e denotes the agreement expected by chance. Specifically, p_e is computed from the marginal distributions of the binary confusion matrix as $p_e = [(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN}) + (\mathrm{TN} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})]/N^2$. Since HSI-CD under limited supervision may suffer from class imbalance, $\kappa \times 100$ and $F1$ are particularly informative because they better reflect the balance between false alarms and missed detections.

B. Experimental Settings and Comparative Analysis

All experiments are implemented in PyTorch 2.11.0 with CUDA 12.8 and conducted on Ubuntu 24.04 with an AMD Ryzen 9 5950X CPU, 128 GB RAM, and an NVIDIA RTX 4070 Ti SUPER GPU with 16 GB memory. Following the protocol for limited samples, 0.1% of labeled pixels are randomly selected from each class (i.e., changed and unchanged) for training. No validation set is used. The selected training pixels are excluded from metric computation, and the remaining labeled pixels are used for quantitative evaluation. Prediction maps for the entire scene are generated for qualitative comparison. This evaluation protocol is applied to all trainable methods for fair comparison. The hyperparameters of all trainable methods are tuned under the same 0.1% protocol, and the best-performing configuration for each method is used for evaluation. To mitigate the effect of random sampling, we conduct multiple independent runs using different random seeds and report the mean and standard deviation.

For MSDistillCD, the maximum input patch size is set to $P = 5$, which corresponds to the main comparison setting

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TABLE I

DETAILED SUMMARY OF THE DATASETS USED IN THE EXPERIMENTS. t_1 AND t_2 DENOTE THE ACQUISITION DATES OF THE FIRST AND SECOND HYPERSPECTRAL IMAGES, RESPECTIVELY

Dataset	Scene location	Sensor	t_1	t_2	Size	Raw/used bands	Spatial/spectral resolution	Changed/ unchanged
River	River area, Jiangsu, China	EO-1 Hyperion	2013-05-03	2013-12-31	463×241	242/198	30 m/about 10 nm	9,698/ 101,885
Farmland	Farmland near Yancheng, Jiangsu, China	EO-1 Hyperion	2006-05-03	2007-04-23	450×140	242/155	30 m/about 10 nm	18,277/ 44,723
USA	Agricultural area near Hermiston, Oregon, USA	EO-1 Hyperion	2004-05-01	2007-05-08	307×241	242/154	30 m/about 10 nm	16,676/ 57,311

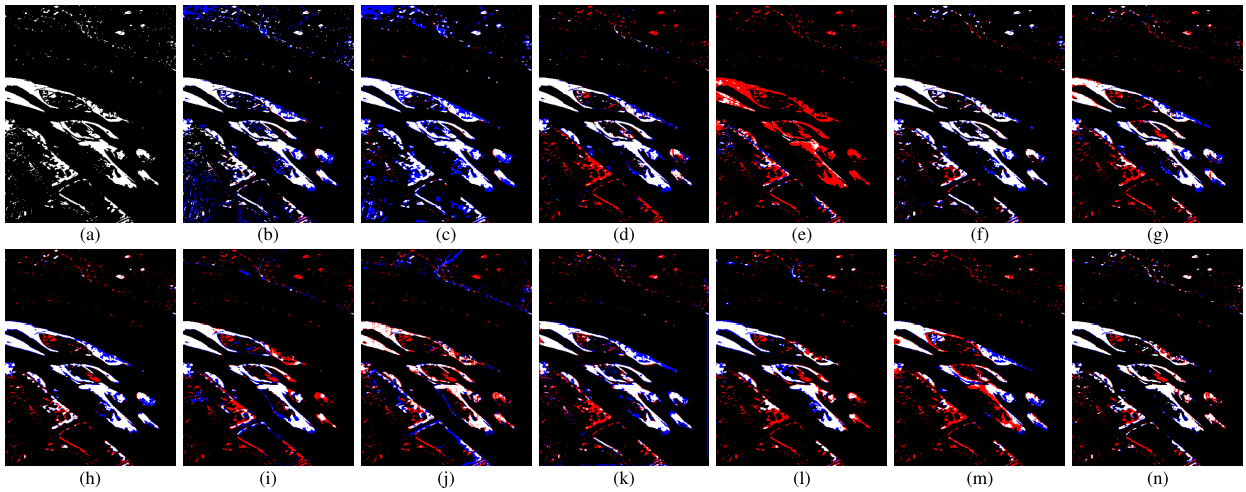


Fig. 5. Qualitative comparison of change detection results on the river dataset under the 0.1% training setting. (a) GT. (b) CVA. (c) PCAKmeans. (d) SVM. (e) RF. (f) GTMSiam. (g) AIWSEN. (h) SpectralKAN. (i) GLAFormer. (j) BTCDNet. (k) SFIEET. (l) DCENet. (m) MSCSCNet. (n) MSDistillCD (Ours). In each map, white and black denote correctly detected changed and unchanged pixels, respectively, while red and blue denote false negatives and false positives, respectively.

$\mathcal{S} = \{3, 5\}$ in our implementation. The backbone uses 64 base channels and three stacked convolutional blocks. The temperature parameter in the self-distillation module is set to $T = 2.0$, and the distillation weight is set to $\alpha = 1.0$. The model is trained for 100 epochs with a batch size of 32 using the SGD optimizer, where the initial learning rate, momentum, and weight decay are set to 3×10^{-4}, 0.9, and 0, respectively. A multistep learning rate schedule is employed, with the learning rate decayed by a factor of 0.1 at epochs 35 and 70. In addition, the primary branch is optimized with the cross-entropy loss defined in Section II.

We compare the proposed MSDistillCD with 12 representative comparable methods, including CVA [8], PCAKmeans [10], SVM [34], RF [35], GTMSiam [16], AIWSEN [36], SpectralKAN [37], GLAFormer [21], BTCDNet [38], SFIEET [39], DCENet [28], and MSCSCNet [29]. All trainable methods are evaluated under the same 0.1% low-label setting. CVA and PCAKmeans do not require training, but they are tested on the same preprocessed datasets and evaluated using the same protocol to ensure fairness. Quantitative results are reported for all compared methods, and representative qualitative maps are also provided.

Table II reports the quantitative results under the 0.1% training setting. The comparison includes traditional approaches, supervised deep baselines tuned for this protocol with extremely limited labeled data, and two recent semi-supervised HSI-CD baselines. MSDistillCD achieves the best OA, $\kappa \times 100$, and $F1$ on all three datasets. On Farmland, it reaches an OA of 96.82%, a $\kappa \times 100$ of 92.40%, and an $F1$ of 94.66%. On River, it obtains an OA of 95.32%, a $\kappa \times 100$ of 68.37%, and an $F1$ of 70.88%. On USA, it achieves an OA of 93.56%, a $\kappa \times 100$ of 81.05%, and an $F1$ of 85.14%. These results demonstrate the effectiveness of the proposed multiscale self-distillation design under extremely limited labeled data.

Figs. 5–7 provide representative qualitative comparisons on River, Farmland, and USA. The visual results are consistent with the quantitative observations in Table II. Across all three datasets, MSDistillCD produces compact change regions with few isolated false alarms and missed detections around complex boundaries. These results indicate that the proposed multiscale representation and self-distillation strategy improve spatial consistency and robustness under extremely limited labeled data.

C. Ablation Studies

1) Spatially Disjoint Row Partition: To evaluate within-scene spatial generalization under extremely limited supervision, we conduct a region-based spatially disjoint

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TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT METHODS ON FARMLAND, RIVER, AND USA DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Methods	OA (%)	$\kappa \times 100$	F1 (%)	P (%)	R (%)
Farmland	MSDistillCD	**96.82±0.44**	**92.40±0.98**	**94.66±0.65**	92.54±2.06	96.92±1.15
	CVA	95.68±0.00	89.60±0.00	92.66±0.00	91.24±0.00	94.12±0.00
	PCAKmeans	94.36±0.00	86.31±0.00	90.28±0.00	90.21±0.00	90.36±0.00
	SVM	95.81±0.61	89.81±1.44	92.75±1.01	**93.32±2.55**	92.28±2.19
	RF	95.42±1.47	88.99±3.50	92.24±2.46	90.92±3.77	93.78±3.70
	GTMSiam	96.01±0.74	90.36±1.75	93.17±1.23	92.67±3.31	93.88±3.35
	AIWSEN	94.69±1.27	87.68±2.71	91.51±1.76	85.93±3.74	**98.01±1.16**
	SpectralKAN	95.97±0.54	90.32±1.16	93.17±0.77	91.94±3.60	94.67±3.22
	GLAFormer	94.97±0.95	88.06±1.98	91.65±1.27	89.00±4.39	94.73±3.06
	BTCDNet	95.23±0.60	88.64±1.29	92.04±0.85	89.54±2.93	94.81±2.27
	SFIEET	93.04±0.96	83.63±2.14	88.63±1.45	84.46±3.15	93.40±3.12
	DCENet	93.45±3.30	83.42±9.19	87.85±7.20	91.57±1.66	85.14±11.58
	MSCSCNet	95.22±0.49	88.47±1.08	91.85±0.73	90.94±2.13	92.84±1.56
River	MSDistillCD	**95.32±0.59**	**68.37±3.67**	**70.88±3.46**	78.06±6.77	65.72±6.85
	CVA	92.68±0.00	65.74±0.00	69.54±0.00	54.47±0.00	**96.13±0.00**
	PCAKmeans	91.87±0.01	62.37±0.03	66.58±0.03	51.79±0.04	93.19±0.03
	SVM	94.07±0.93	60.45±4.66	63.62±4.40	69.95±9.14	59.98±8.82
	RF	94.12±0.82	53.88±6.57	56.72±6.40	80.62±10.92	44.84±8.01
	GTMSiam	94.97±0.31	66.77±1.93	69.49±1.80	73.73±3.44	65.96±3.68
	AIWSEN	94.27±0.48	59.50±5.23	62.48±5.26	73.58±6.83	56.00±11.13
	SpectralKAN	94.44±1.71	53.73±28.23	55.75±29.24	73.46±27.64	50.99±28.57
	GLAFormer	91.82±1.68	49.62±6.04	53.95±5.82	56.11±11.50	55.67±12.34
	BTCDNet	91.45±1.99	38.39±11.77	42.72±11.18	54.02±15.90	37.41±13.28
	SFIEET	92.65±0.66	52.55±8.05	56.50±7.93	57.96±4.36	56.40±12.33
	DCENet	93.06±1.29	61.26±3.09	64.99±2.58	59.67±7.39	73.52±8.45
	MSCSCNet	93.56±0.53	44.92±10.47	47.65±10.80	**81.47±8.79**	35.22±12.07
USA	MSDistillCD	**93.56±1.31**	**81.05±3.58**	**85.14±2.77**	89.42±6.38	81.76±5.09
	CVA	92.00±0.00	74.13±0.00	78.83±0.00	97.72±0.00	66.06±0.00
	PCAKmeans	92.08±0.00	74.28±0.00	78.92±0.00	**98.65±0.00**	65.76±0.01
	SVM	92.08±1.09	76.75±2.71	81.78±2.08	85.77±6.54	78.80±5.58
	RF	88.02±2.05	63.53±5.88	70.95±4.75	79.08±7.83	65.13±7.05
	GTMSiam	92.65±2.14	78.68±6.32	83.37±5.01	85.42±6.58	81.97±7.49
	AIWSEN	92.83±1.32	79.36±4.05	83.97±3.26	84.63±4.44	**83.72±6.04**
	SpectralKAN	93.24±2.01	79.71±7.25	83.94±6.22	88.83±2.49	80.16±9.79
	GLAFormer	88.45±2.51	68.21±6.03	75.75±4.36	72.74±7.17	79.42±3.69
	BTCDNet	88.22±2.33	66.60±5.87	74.19±4.44	74.09±6.95	74.98±6.49
	SFIEET	86.39±1.50	61.53±3.75	70.32±2.96	69.58±4.79	71.66±6.24
	DCENet	92.21±1.75	77.71±4.76	82.73±3.63	83.07±5.02	82.63±4.33
	MSCSCNet	92.07±1.43	75.58±4.90	80.48±4.11	89.93±2.60	73.01±6.19

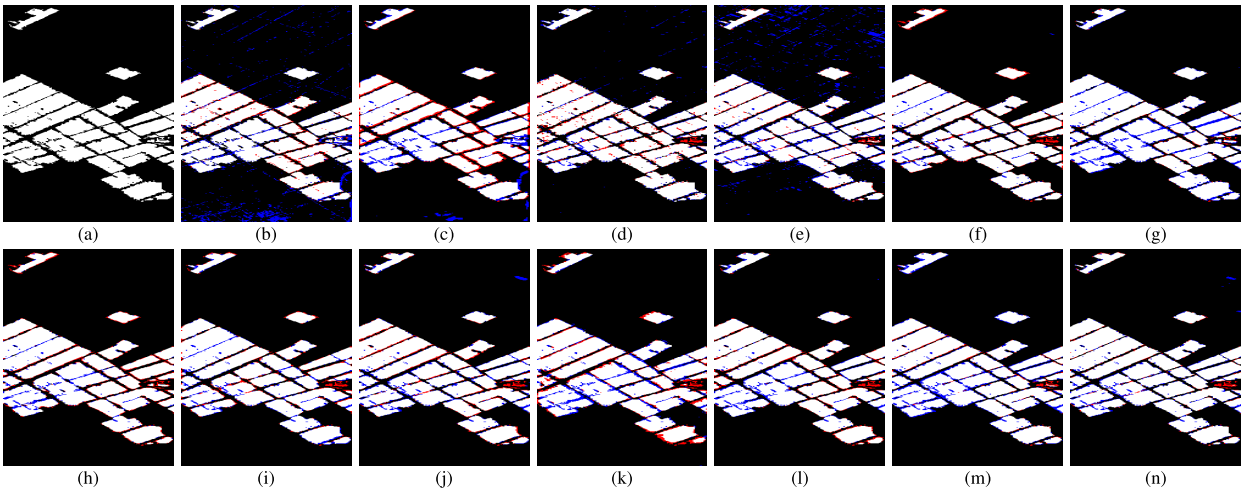


Fig. 6. Qualitative comparison of change detection results on the farmland dataset under the 0.1% training setting. (a) GT. (b) CVA. (c) PCAKmeans. (d) SVM. (e) RF. (f) GTMSiam. (g) AIWSEN. (h) SpectralKAN. (i) GLAFormer. (j) BTCDNet. (k) SFIEET. (l) DCENet. (m) MSCSCNet. (n) MSDistillCD (Ours). In each map, white and black denote correctly detected changed and unchanged pixels, respectively, while red and blue denote false negatives and false positives, respectively.

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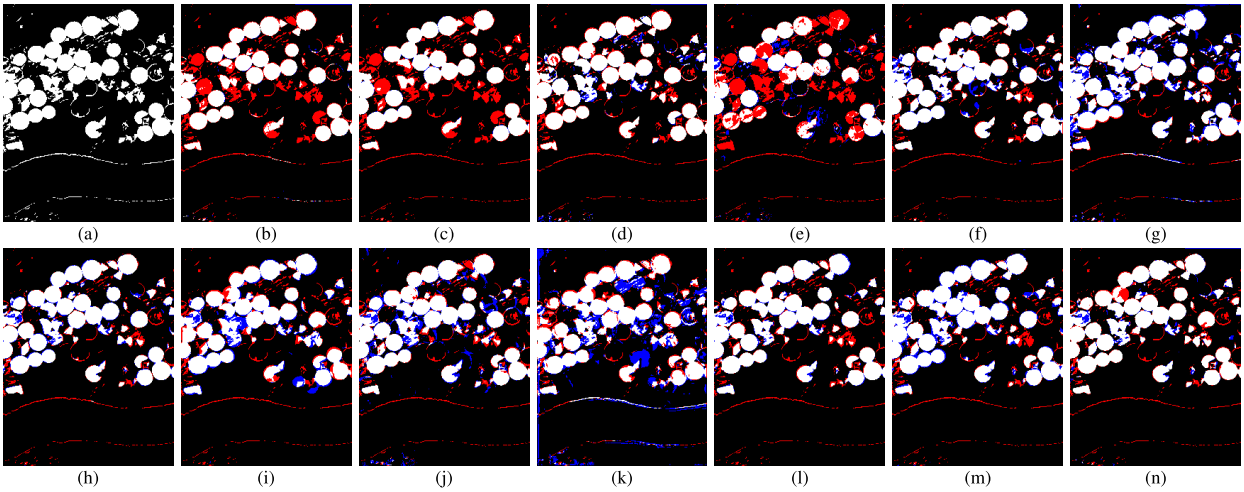


Fig. 7. Qualitative comparison of change detection results on the USA dataset under the 0.1% training setting. (a) GT. (b) CVA. (c) PCAKmeans. (d) SVM. (e) RF. (f) GTMSiam. (g) AIWSEN. (h) SpectralKAN. (i) GLAFormer. (j) BTCDNet. (k) SFIEET. (l) DCENet. (m) MSCSCNet. (n) MSDistillCD (Ours). In each map, white and black denote correctly detected changed and unchanged pixels, respectively, while red and blue denote false negatives and false positives, respectively.

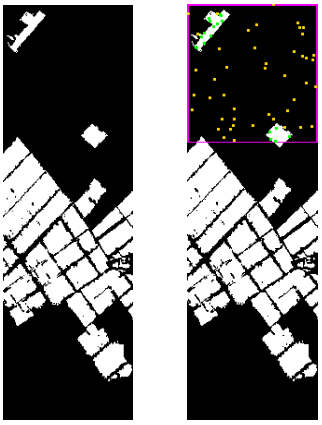


Fig. 8. Illustration of the spatially disjoint row partition on the Farmland dataset. Green and yellow denote the 18 changed and 45 unchanged training pixels, respectively.

TABLE III

RESULTS UNDER THE SPATIALLY DISJOINT ROW PARTITION ON FARMLAND AT THE 0.1% LABELED RATIO. VALUES ARE MEAN ± STANDARD DEVIATION; BEST RESULTS ARE IN BOLD

Method	OA (%)	$\kappa \times 100$	F1 (%)
MSDistillCD	**94.55±0.88**	**88.85±1.78**	**93.59±0.99**
SpectralKAN	83.68±11.74	63.89±27.35	73.90±22.22
GTMSiam	88.23±9.49	74.63±22.17	82.81±18.15
BTCDNet	93.17±0.90	86.10±1.71	92.08±0.82
GLAFormer	89.10±0.98	78.29±1.83	88.20±0.85
AIWSEN	86.15±9.25	70.67±21.77	80.99±17.81
SFIEET	88.67±1.12	77.39±2.10	87.66±0.99

experiment on the Farmland dataset under the 0.1% setting. Labeled samples are selected from a candidate annotation region, whereas evaluation is performed in a spatially nonoverlapping test region. All compared methods use the same sampled training pixels in each run. The detailed partition is shown in Fig. 8.

Compared with the random sampling results in Table II, all methods show a clear overall performance decrease under the spatially disjoint row partition, as shown in Table III. Nevertheless, MSDistillCD retains the highest OA, $\kappa \times 100$, and $F1$. It exceeds the strongest competing result by 1.38, 2.75, and 1.51 percentage points in OA, $\kappa \times 100$, and $F1$, respectively. These results demonstrate that MSDistillCD maintains its advantage when the annotation and test regions are spatially separated.

2) Component Ablation: To evaluate the contributions of MSFEM and SDCM, we conduct ablation experiments by comparing the full MSDistillCD model with two reduced variants. The variant using MSFEM but removing SDCM corresponds to multiscale training with hard labels and without self-distillation. The detailed results are reported in Table IV.

As shown in Table IV, the complete MSDistillCD model achieves the highest OA, $\kappa \times 100$, $F1$, and precision on River, Farmland, and USA. Adding SDCM to the MSFEM variant consistently improves OA, $\kappa \times 100$, $F1$, and precision on all three datasets. The gains in OA, $\kappa \times 100$, and $F1$ are 0.53, 1.78, and 1.48 percentage points on River; 0.86, 1.89, and 1.26 points on Farmland; and 0.46, 0.50, and 0.12 points on USA, respectively. These results verify the effectiveness of cross-scale regularization and the complementary design of MSFEM and SDCM under extremely limited labeled data.

3) Fine-Grained Ablation: We further conduct a detailed ablation on the Farmland dataset to isolate the effects of the difference-aware input, aggregation strategy, backbone sharing, and self-distillation direction. As reported in Table V, using only the raw temporal pair or only the absolute difference is less effective than combining raw spectral information with explicit difference cues. The proposed input representation achieves the best OA, $\kappa \times 100$, and $F1$ among the selected input variants. In addition, dual aggregation outperforms center aggregation alone and max aggregation alone, and the shared backbone achieves

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TABLE IV

ABLATION RESULTS OF DIFFERENT VARIANTS ON THE RIVER, FARMLAND, AND USA DATASETS. THE RESULTS ARE REPORTED AS MEAN ± STANDARD DEVIATION OVER 10 RUNS

Dataset	MSFEM	SDCM	OA (%)	$\kappa \times 100$	F1 (%)	P (%)	R (%)
River	×	×	94.88 ± 0.46	67.10 ± 3.20	69.86 ± 3.14	72.41 ± 6.41	**68.75 ± 7.70**
	✓	×	94.79 ± 0.73	66.59 ± 3.70	69.40 ± 3.49	72.38 ± 7.69	68.25 ± 8.53
	✓	✓	**95.32 ± 0.59**	**68.37 ± 3.67**	**70.88 ± 3.46**	**78.06 ± 6.77**	65.72 ± 6.85
Farmland	×	×	96.08 ± 0.57	90.72 ± 1.24	93.52 ± 0.82	90.02 ± 2.30	97.37 ± 1.03
	✓	×	95.96 ± 0.57	90.51 ± 1.26	93.40 ± 0.84	89.00 ± 1.89	**98.29 ± 0.48**
	✓	✓	**96.82 ± 0.44**	**92.40 ± 0.98**	**94.66 ± 0.65**	**92.54 ± 2.06**	96.92 ± 1.15
USA	×	×	92.92 ± 1.56	79.82 ± 3.96	84.38 ± 2.97	84.86 ± 6.18	84.46 ± 4.85
	✓	×	93.10 ± 1.42	80.55 ± 3.68	85.02 ± 2.75	83.92 ± 5.61	**86.54 ± 3.87**
	✓	✓	**93.56 ± 1.31**	**81.05 ± 3.58**	**85.14 ± 2.77**	**89.42 ± 6.38**	81.76 ± 5.09

TABLE V

DETAILED ABLATION ON FARMLAND WITH 0.1% TRAINING SAMPLES. VALUES ARE MEAN ± STANDARD DEVIATION OVER TEN RUNS

(a) Difference-aware input

Variant	OA (%)	$\kappa \times 100$	F1 (%)
Raw pair only	96.44±0.26	91.34±0.68	93.85±0.51
Absolute difference only	95.84±0.63	89.87±1.62	92.78±1.19
Raw + absolute difference	96.53±0.42	91.54±1.04	93.98±0.76
Ours	**96.82±0.44**	**92.40±0.98**	**94.66±0.65**

(b) Aggregation strategy

Variant	OA (%)	$\kappa \times 100$	F1 (%)
Center aggregation	96.39±0.37	91.34±0.81	93.90±0.54
Max aggregation	96.47±0.37	91.39±0.97	93.87±0.73
Ours	**96.82±0.44**	**92.40±0.98**	**94.66±0.65**

(c) Backbone sharing

Variant	OA (%)	$\kappa \times 100$	F1 (%)
Independent backbones	96.40±0.40	91.39±0.91	93.94±0.62
Ours	**96.82±0.44**	**92.40±0.98**	**94.66±0.65**

(d) Distillation direction

Variant	OA (%)	$\kappa \times 100$	F1 (%)
Large-to-small ($5 \rightarrow 3$)	96.52±0.39	91.48±1.00	93.91±0.73
Ours	**96.82±0.44**	**92.40±0.98**	**94.66±0.65**

higher OA, $\kappa \times 100$, and $F1$ than independent backbones on this representative dataset. For the comparison of distillation directions, the involved scales are 3 and 5, which are the default scales used in the main experiments. Specifically, small-to-large denotes distillation from scale 3 to scale 5, whereas large-to-small denotes the reverse direction from scale 5 to scale 3. Under this setting, the proposed small-to-large direction achieves higher OA, $\kappa \times 100$, and $F1$ than the reverse large-to-small direction.

Overall, the ablation study verifies the contributions of the proposed components. MSFEM supplies center-aligned contextual views, and SDCM regularizes their predictions across scales. Their combination consistently achieves the strongest OA, $\kappa \times 100$, and $F1$ results on all three datasets.

D. Parameter Sensitivity and Efficiency Analysis

This section analyzes the behavior of MSDistillCD under different annotation budgets and parameter settings in the regime with limited samples, and further compares the practical computational efficiency of different deep-learning methods under their tuned settings. In particular, we compare the OA, $\kappa \times 100$, and $F1$ values of different methods on the Farmland dataset under training ratios from 0.05% to 0.20%, evaluate the sensitivity of the distillation weight α, the temperature T, and the scale setting $\mathcal{S}$, and summarize the efficiency statistics on the River dataset.

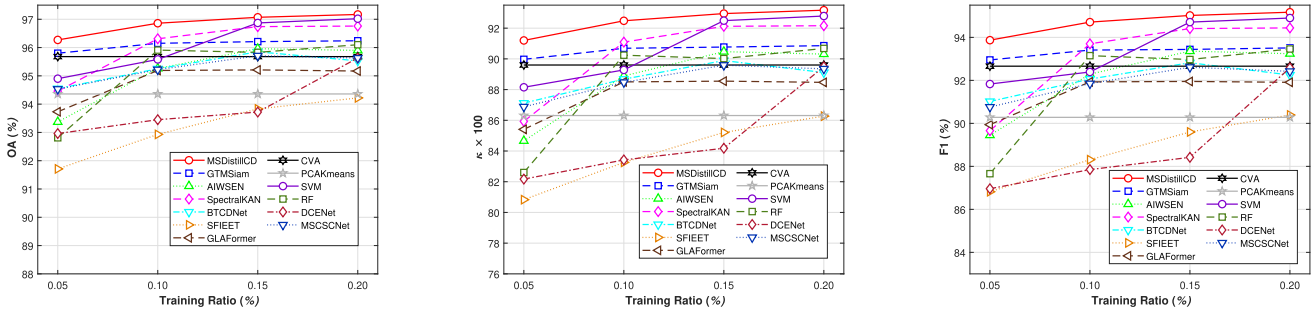
For all trainable methods, the ratio denotes the proportion of labeled pixels. The semi-supervised methods use the same random seeds and labeled splits as the supervised methods. Following the default settings in their original papers, the unlabeled pools are fixed at 20% for MSCSCNet and 10% for DCENet. Thus, their curves measure sensitivity to the scarcity of labeled data; no test labels are used for training. Because CVA and PCAKmeans require no labels, their fixed results are shown as horizontal references.

As shown in Fig. 9, MSDistillCD consistently achieves the highest mean OA, $\kappa \times 100$, and $F1$ over the examined labeled ratios from 0.05% to 0.20%. Its lead is most pronounced under the smallest annotation budget and remains consistent across all three metrics as the number of labeled samples increases. This result demonstrates the robustness of MSDistillCD under different annotation budgets. The semi-supervised curves should be interpreted with the fixed unlabeled pools described above; they therefore measure sensitivity to the scarcity of labeled data rather than to simultaneous changes in labeled and unlabeled data.

Fig. 10(a) illustrates the effect of the distillation weight α on OA. The OA values remain stable as α varies from 0 to 1 across the three datasets, demonstrating that MSDistillCD is robust to the choice of this hyperparameter. The default setting $\alpha = 1$ provides effective self-distillation regularization and lies within a stable operating region across datasets.

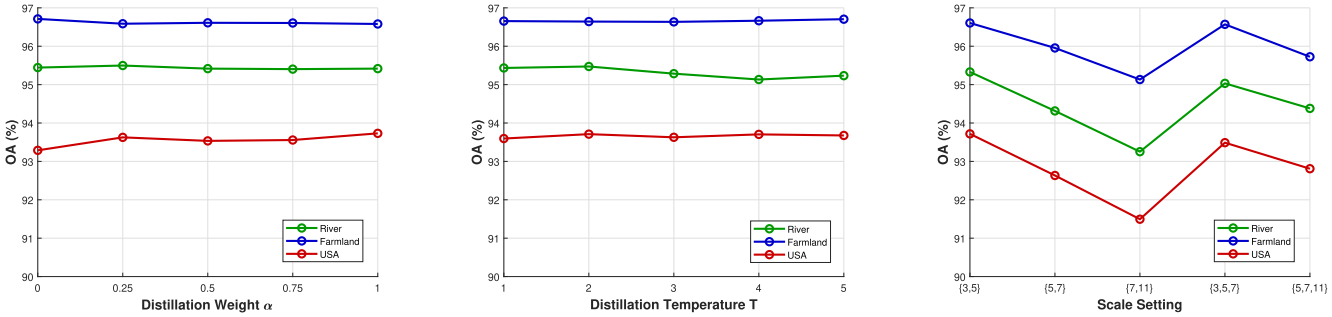
Fig. 10(b) shows the influence of the distillation temperature T on OA. The OA values remain stable over the tested temperature range across all three datasets. The default setting $T = 2$ provides effective soft supervision and lies

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(a) (b) (c)

Fig. 9. Comparison of different methods on Farmland under labeled training ratios from 0.05% to 0.20%. (a) OA. (b) $\kappa \times 100$. (c) $F1$. Curves show means over five runs. For MSCSCNet and DCENet, the ratio denotes labeled samples only; the unlabeled pool is fixed across ratios.



(a) (b) (c)

Fig. 10. Parameter sensitivity analysis of MSDistillCD using OA. (a)–(c) OA scores for each dataset with respect to the distillation weight α, the temperature T, and the scale setting $\mathcal{S}$, respectively.

within a stable operating region, demonstrating the robustness of MSDistillCD to the choice of temperature.

Fig. 10(c) presents the effect of different scale settings on OA. For this experiment, the maximum patch size is set to $P = \max \mathcal{S}$ for each tested scale set so that all evaluated scales are valid. It can be clearly observed that the default setting $\mathcal{S} = \{3, 5\}$ achieves the best OA on all three datasets, demonstrating that a compact design with two scales is sufficient to capture complementary contextual information under the setting with extremely limited samples. In contrast, introducing larger receptive fields, such as $\mathcal{S} = \{5, 7\}$, $\mathcal{S} = \{7, 11\}$, $\mathcal{S} = \{3, 5, 7\}$, or $\mathcal{S} = \{5, 7, 11\}$, does not bring further gains and instead leads to consistent performance degradation, especially on River and USA. This result suggests that overly large neighborhoods may introduce redundant or noisy contextual information and make optimization more difficult when only a very limited number of labeled pixels are available. In addition, larger scale settings also incur higher computational cost. Therefore, $\mathcal{S} = \{3, 5\}$ provides the best balance between effectiveness and efficiency and is used as the default multiscale configuration.

For the efficiency comparison on River, all experiments were conducted on the same hardware platform under the 0.1% training-ratio protocol. Each supervised deep-learning method was evaluated under its tuned setting. Table VI reports parameter count, peak inference GPU memory, full-image inference time, throughput, and total training time.

TABLE VI

EFFICIENCY COMPARISON OF THE LISTED SUPERVISED DEEP LEARNING METHODS ON RIVER UNDER THEIR TUNED SETTINGS

Methods	Params (K)	Inference GPU Memory (MB)	Infer. Time (s)	Pixels/s	Train Time (s)
MSDistillCD	530.82	33.73	**7.40**	**15062.74**	**2.15**
GTMSiam	63.17	**21.22**	24.55	4544.71	9.17
AIWSEN	12504.76	180.97	67.20	1660.35	22.64
SpectralKAN	**3.54**	44.30	13.15	8479.23	5.48
GLAFormer	12646.83	235.54	84.05	1327.54	217.89
BTCDNet	1594.58	44.88	61.86	1803.79	33.44
SFIEET	1885.57	67.64	38.14	2924.89	13.36

As shown in Table VI, MSDistillCD achieves the shortest inference time (7.4079 s), the highest throughput (15 062.74 pixels/s), and the shortest training time (2.153 s). Its full-image inference and training are approximately 1.78× and 2.55× faster than those of SpectralKAN, respectively. MSDistillCD uses 33.73 MB of inference GPU memory and contains 530.82 K parameters, demonstrating a favorable balance among model compactness, memory consumption, and execution speed.

IV. CONCLUSION

This article presents MSDistillCD for HSI-CD with extremely limited labeled samples. The framework consists of MSFEM and SDCM. MSFEM constructs center-aligned multiscale inputs by integrating bitemporal spectral information with signed and absolute differences, and uses a shared

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lightweight CNN with dual global aggregation to extract discriminative features. SDCM employs the smallest-scale branch as the primary branch to guide the larger-scale auxiliary branches during training, enabling multiscale consistency learning while retaining single-branch inference. Experiments on the River, Farmland, and USA datasets demonstrate that MSDistillCD achieves the best OA, $\kappa \times 100$, and $F1$ among the compared methods under the 0.1% training setting. The ablation and efficiency results further verify the effectiveness and computational efficiency of the proposed framework.

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